

Titanic Dataset Data Preprocessing Report

Objective: To perform data preprocessing on the Titanic dataset using Python libraries such as Pandas and NumPy by handling missing values, removing duplicates, handling outliers, and preparing clean data for analysis.

Tools Used

- Python
- Pandas
- NumPy
- Google Colab

Dataset Used

Titanic Dataset

Preprocessing Steps Performed

1. Loaded the Titanic dataset using Pandas.
2. Checked dataset information and missing values.
3. Handled missing values in Age and Fare columns.
4. Removed duplicate rows.
5. Handled outliers using the IQR method.
6. Normalized numerical features.
7. Converted categorical columns into numerical format using encoding.
8. Saved the cleaned dataset.

Before Data Cleaning

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	86
SibSp	0
Parch	0
Ticket	0
Fare	1
Cabin	327
Embarked	0

dtype: int64

After Data Cleaning

```
> df.isnull().sum()
..
PassengerId  0
Survived     0
Pclass       0
Name         0
Sex          0
Age         0
SibSp        0
Parch        0
Ticket       0
Fare         0
Embarked_C   0
Embarked_Q   0
Embarked_S   0
dtype: int64
```

Final Outcome

The dataset was successfully cleaned and preprocessed. Missing values were handled, duplicates were removed, categorical data was encoded, and the dataset was prepared for further analysis and machine learning tasks.

Files Included

File Name	Description
titanicdataset.csv	Original dataset
Cleaned_Titanic_Dataset.csv	Cleaned dataset
Data_preprocessing.ipynb	Google colab
README.md	Project documentation
Report.docx	Project report

Conclusion

This project helped in understanding the basics of data preprocessing including handling missing values, duplicate removal, outlier treatment, normalization, and categorical encoding.